LAB TASK # 01

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Python Setup, Fundamentals, and Imports

COURSE	Digital Image Processing
SEMESTER	5th
INSTRUCTOR	Engr. Fahad Nisar
ASSIGNED	16 October 2026
DUE	23 October 2026

01 / Python installation and setup

Python and pip were verified in cmd.

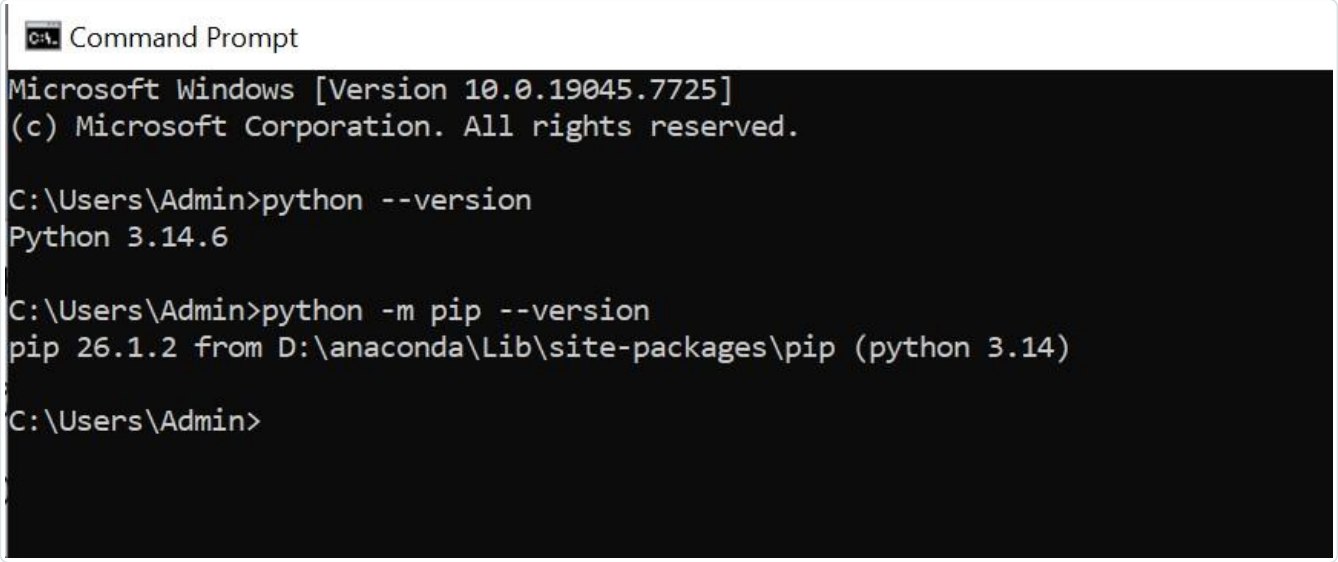


Figure 1. Command Prompt: Python 3.14.6 and pip 26.1.2.

Visual Studio Code installed

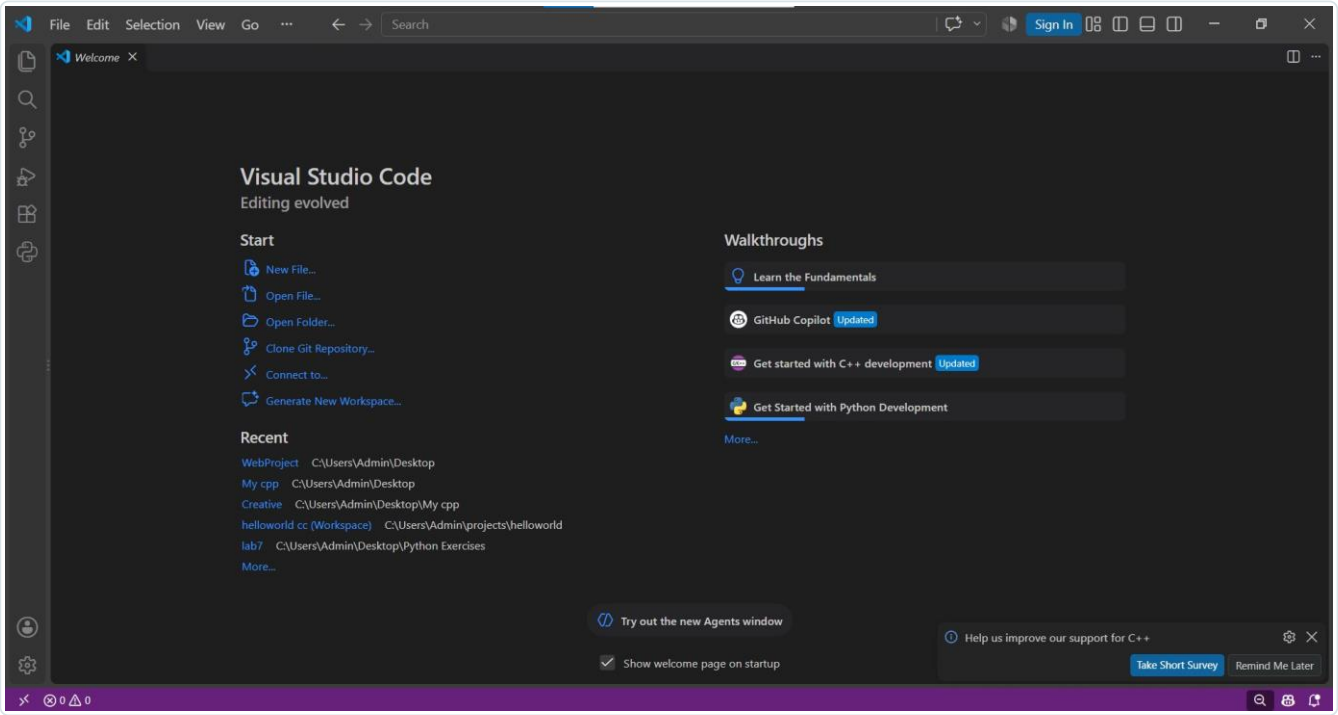


Figure 2. Visual Studio Code application opened successfully.

Verified environment Windows 10 | Python 3.14.6 | pip 26.1.2

02 / VS Code Python configuration

Microsoft Python extension

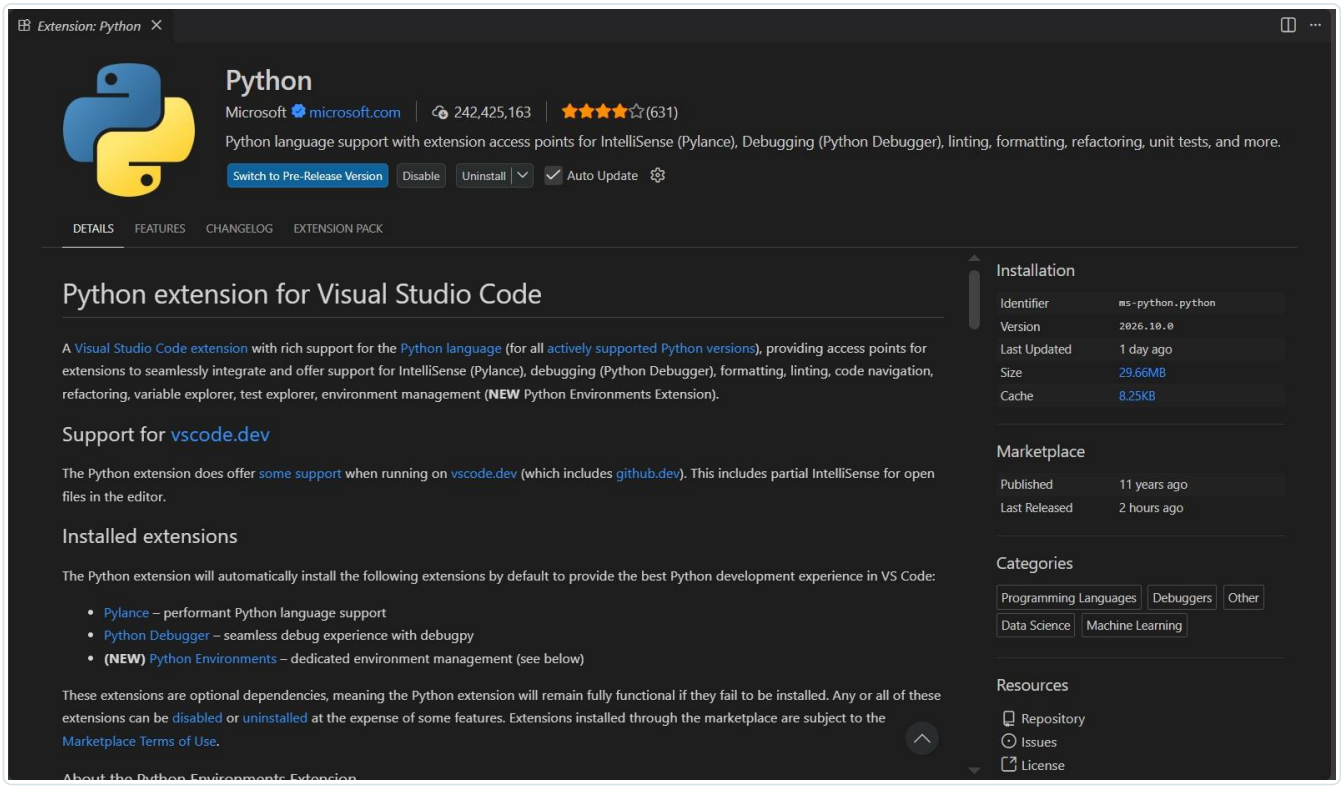


Figure 3. Python extension by Microsoft installed in VS Code.

Selected Python interpreter

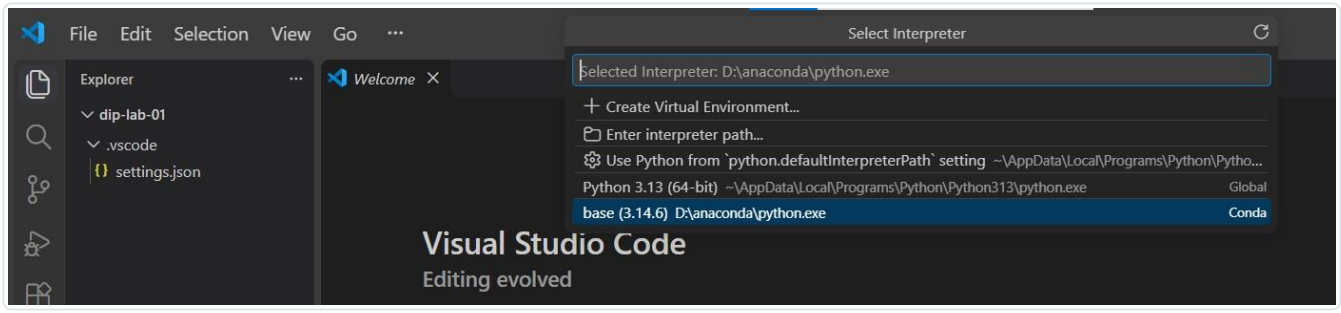


Figure 4. Anaconda base (Python 3.14.6), D:\anaconda\python.exe.

The Python extension enables Python editing and execution in VS Code. The base Anaconda interpreter was selected so the editor runs the installed Python environment

03 / First Python program

The file hello.py was saved in the dip-lab-01 directory and executed successfully in both VS Code and Command Prompt.

Source code and VS Code execution

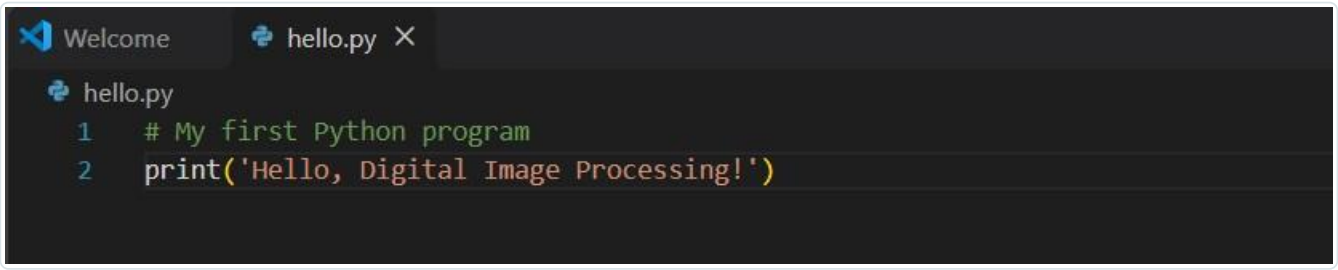


Figure 5a. hello.py source code.



Figure 5b. Successful execution in the VS Code terminal.

Command Prompt verification

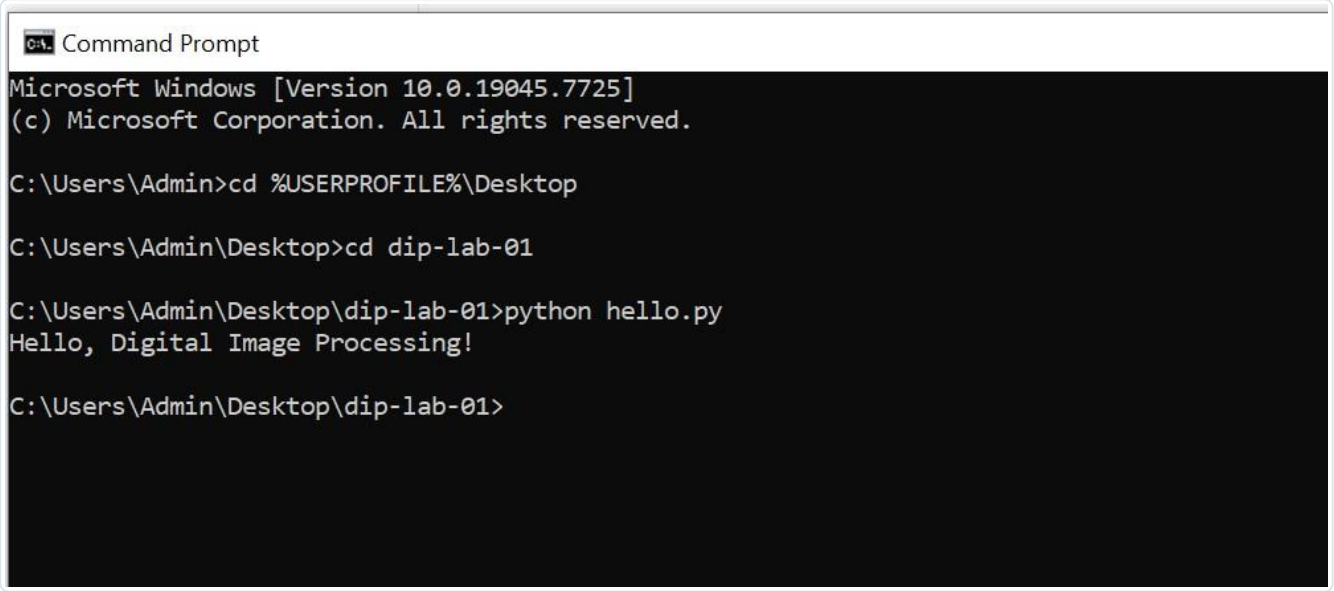


Figure 6. Running python hello.py from the project folder.

EXPECTED AND OBSERVED OUTPUT

```
Hello, Digital Image Processing!
```

04 / Python fundamentals: exercises 1-2

The basics.py script includes labeled sections, four-space indentation and printed results for all six programming exercises.

Variables, data types and conditional statements

basics.py > ...

```python
# Exercise 1: Variables and Data Types
# ========================================

print("\n--- Exercise 1: Variables and Data Types ---")

width = 640
height = 480
image_name = "sample.png"
is_color = True

print("Width:", width, "| Type:", type(width))
print("Height:", height, "| Type:", type(height))
print("Image Name:", image_name, "| Type:", type(image_name))
print("Is Color:", is_color, "| Type:", type(is_color))

total_pixels = width * height
print("Total Pixels:", total_pixels)


# Exercise 2: if, elif, and else
# ========================================

print("\n--- Exercise 2: Conditions ---")

intensities = [-1, 0, 127, 128, 255, 256]

for intensity in intensities:
    if intensity < 0 or intensity > 255:
        label = "Invalid"
    elif intensity >= 128:
        label = "Bright pixel"
    else:
        label = "Dark pixel"

    print("Intensity:", intensity, "->", label)
```

Figure 7. basics.py source for Exercises 1 and 2.

CHECKED RESULTS

Exercise 1	640 x 480 = 307200 total pixels; correct Python data types.
Exercise 2	-1, 256: Invalid | 0, 127: Dark | 128, 255: Bright

05 / Python fundamentals: exercises 3-4

For / while loops and functions

```python
# Exercise 3: for and while Loops
# ========================================

print("\n--- Exercise 3: Loops ---")

pixels = [20, 130, 255]

for pixel in pixels:
    if pixel >= 128:
        print(pixel, "-> Bright pixel")
    else:
        print(pixel, "-> Dark pixel")

print("While Loop:")

counter = 0
while counter < 3:
    print(counter)
    counter += 1


# Exercise 4: Functions
# ========================================

print("\n--- Exercise 4: Functions ---")

def is_bright(pixel, threshold=128):
    return pixel >= threshold

print("is_bright(127):", is_bright(127))
print("is_bright(128):", is_bright(128))
print("is_bright(200, 210):", is_bright(200, 210))

```

Figure 8. basics.py source for Exercises 3 and 4.

CHECKED RESULTS

Exercise 3	20: Dark | 130: Bright | 255: Bright
	While loop: 0, 1, 2 (stops before 3)
Exercise 4	is_bright(127) = False; is_bright(128) = True
	is_bright(200, 210) = False

A threshold of 128 classifies a valid grayscale intensity as bright when its value is greater than or equal to the threshold.

06 / Python fundamentals: exercises 5-6

Lists and tuples

```
75  # Exercise 5: Lists and Tuples
76  # ========================================
77
78  print("\n--- Exercise 5: Lists and Tuples ---")
79
80  pixels = [0, 128, 255]
81
82  print("First Pixel:", pixels[0])
83
84  pixels[1] = 150
85  pixels.append(64)
86
87  print("Updated Pixels:", pixels)
88  print("List Length:", len(pixels))
89
90  size = (640, 480)
91  width, height = size
92
93  print("Width:", width)
94  print("Height:", height)
95
96  # Lists are mutable, so their items can be changed.
97  # Tuples are immutable, so their items cannot be changed.
98
```

Figure 9. Modifying a list and unpacking an immutable tuple.

Dictionaries and sets

```python
# Exercise 6: Dictionaries and Sets
# ========================================

print("\n--- Exercise 6: Dictionaries and Sets ---")

image = {"width": 640, "height": 480}

image["mode"] = "RGB"

print("Image Width:", image["width"])
print("Image Mode:", image["mode"])

labels = ["cat", "dog", "cat", "bird"]

unique_labels = set(labels)

print("Unique Labels:", unique_labels)
print("Number of Unique Labels:", len(unique_labels))

```

Figure 10. Image metadata dictionary and unique label set.

07 / Verified output from basics.py

The following terminal screenshots show the outputs of all six exercises, including correct boundary conditions, loop values, collection operations and return values.

Exercises 1-3 output

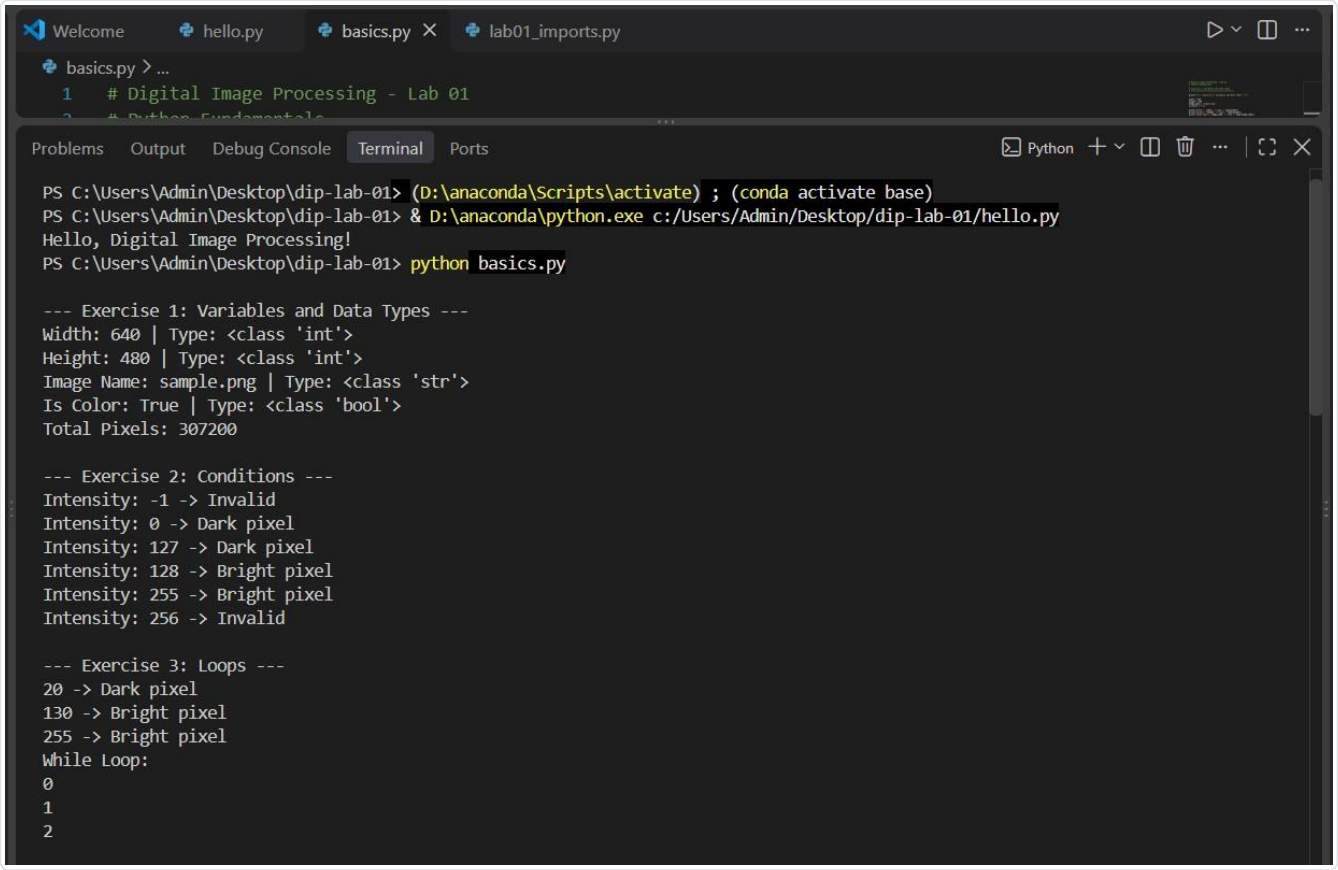


Figure 11. Terminal output: variables, conditions and loops.

Exercises 4-6 output

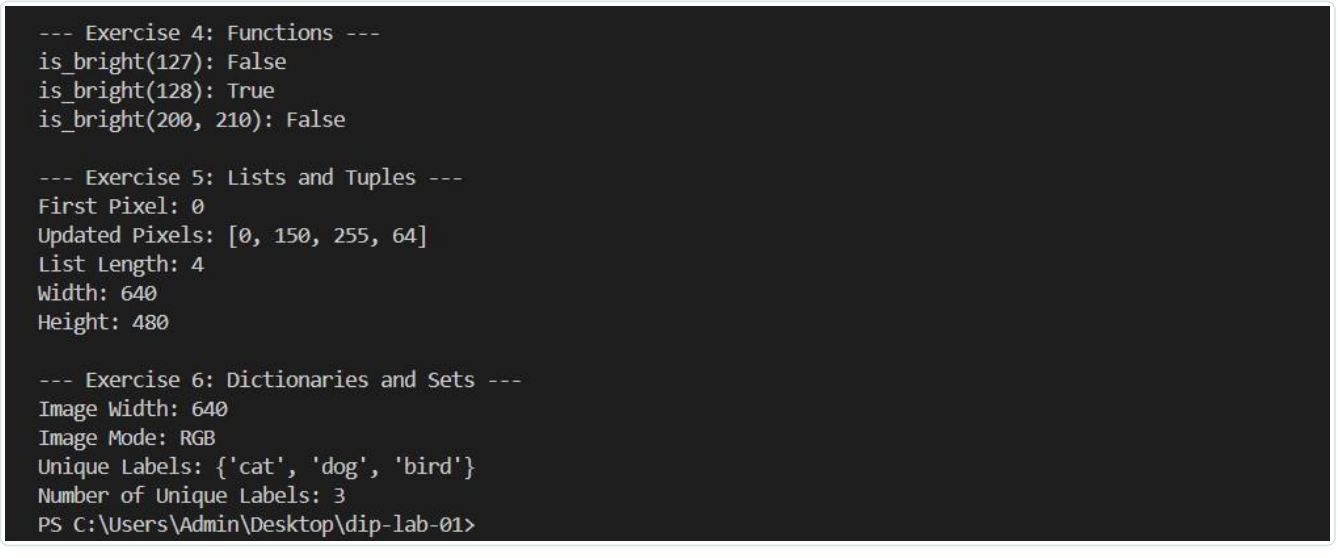


Figure 12. Terminal output: functions, collections and exit to prompt.

Result: All six exercises executed successfully with expected values.

08 / Installation of required packages

The required libraries were confirmed using the Command Prompt command shown below.

```
python -m pip show numpy pillow matplotlib
```

Installed package details

```
C:\Users\Admin>python -m pip show numpy pillow matplotlib
Name: numpy
Version: 2.4.6
Summary: Fundamental package for array computing in Python
Home-page: https://numpy.org
Author: Travis E. Oliphant et al.
Author-email:
License-Expression: BSD-3-Clause AND 0BSD AND MIT AND Zlib AND CC0-1.0
Location: D:\anaconda\Lib\site-packages
Requires:
Required-by: astropy, bokeh, contourpy, datashader, h5py, holoviews, hvplot, ImageIO, imbalanced-learn, ipympl,
sy, pyerfa, PyWavelets, scikit-image, scikit-learn, scipy, seaborn, statsmodels, streamlit, tables, tifffile, x
```

Figure 13a. NumPy 2.4.6 - name, version and installation location.

```
Name: pillow
Version: 12.3.0
Summary: Python Imaging Library (fork)
Home-page: https://python-pillow.github.io
Author:
Author-email: Jeffrey 'Alex' Clark <aclark@aclark.net>
License-Expression: MIT-CMU
Location: D:\anaconda\Lib\site-packages
Requires:
Required-by: bokeh, ImageIO, ipympl, matplotlib, scikit-image, streamlit
---
```

Figure 13b. Pillow 12.3.0 - name, version and installation location.

```
---
Name: matplotlib
Version: 3.11.0
Summary: Python plotting package
Home-page: https://matplotlib.org
Author: John D. Hunter, Michael Droettboom
Author-email: Unknown <matplotlib-users@python.org>
```

Figure 13c. Matplotlib 3.11.0 - name and version.

PURPOSE OF LIBRARIES

NumPy: numerical arrays | Pillow: image handling

Matplotlib: image display and plotting

09 / Import verification and conclusion

Import statements and successful execution

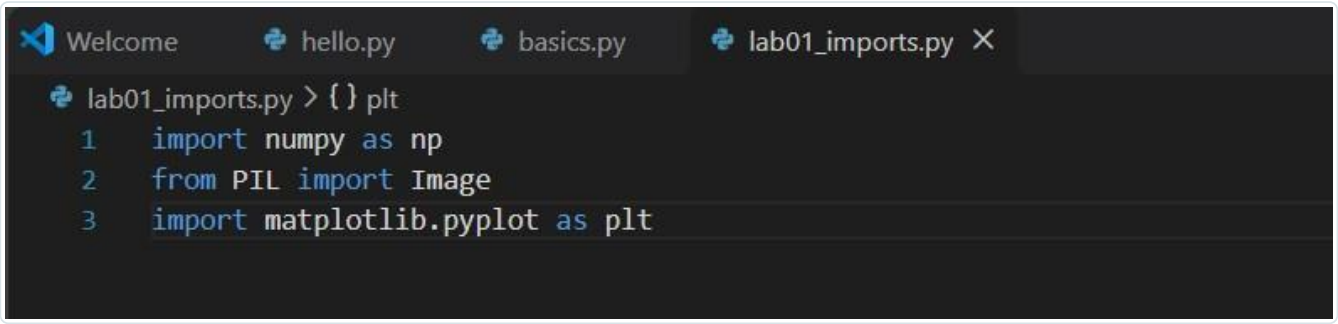


Figure 14a. lab01_imports.py: required import statements.

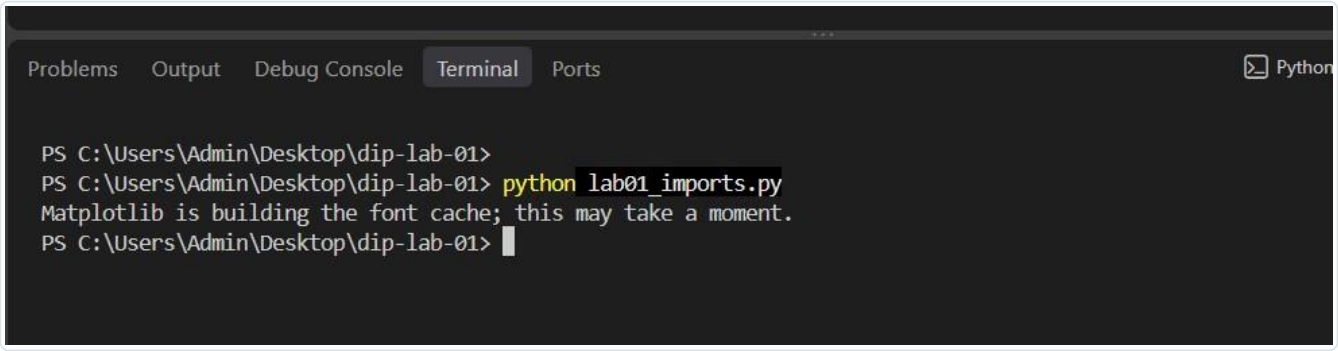


Figure 14b. python lab01_imports.py returned to the prompt without errors.

Python, pip and imports …

Python is the programming language and runtime used to execute scripts such as hello.py and basics.py.

pip is the package installer for Python, used to install and inspect external libraries such as NumPy, Pillow and Matplotlib.

The import statements in lab01_imports.py load these libraries so the program can access numerical arrays, image operations and plotting features which we also observed/discussed in the lab class.

CONCLUSION

Python, pip and VS Code were configured and verified. All six Python fundamentals exercises produced their expected outputs, the three required packages were available, and the import-only script completed without errors.

ACCOMPANYING SUBMISSION FILES

`hello.py` | `basics.py` | `lab01_imports.py`

THANK YOU!
